

Simple RPC Library (SRPC)

SRPC is an experimental RPC framework that uses Python interfaces as its Interface Definition Language. Define a service contract once, generate client and server stubs, implement the service, and invoke remote procedures.

Features

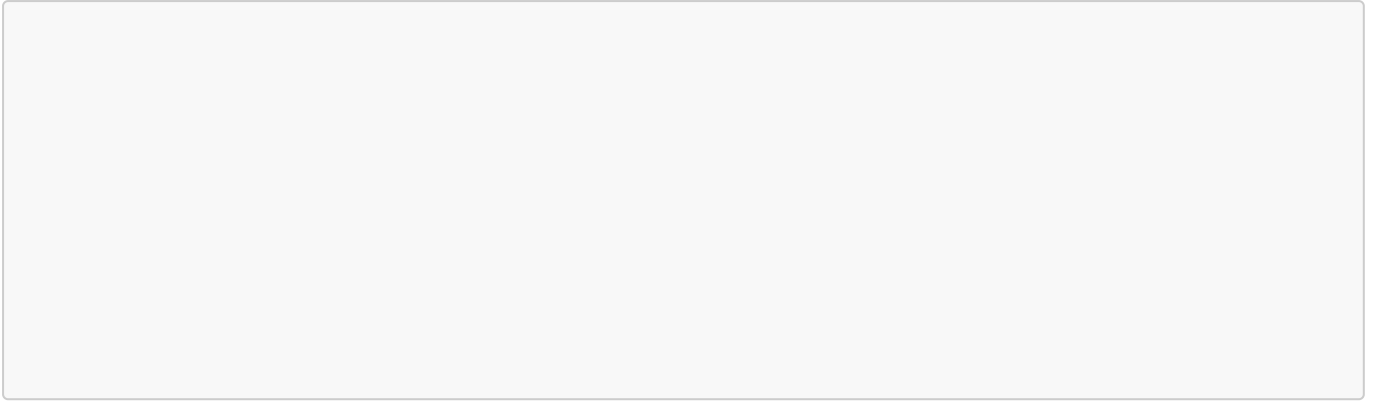
- Python as IDL
- Automatic Stub generation
- TLS support

[!WARNING] Currently, the framework only supports Python language.

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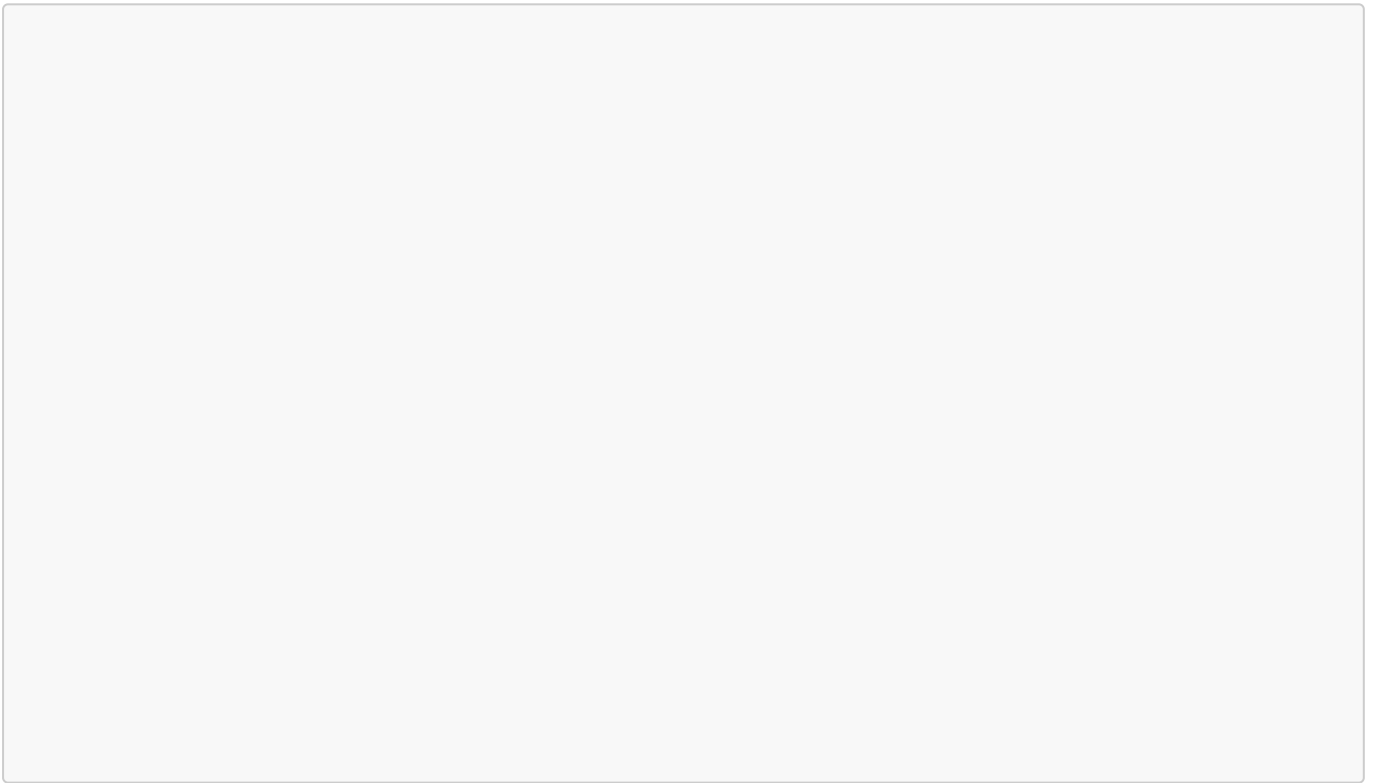
Server side directory structure



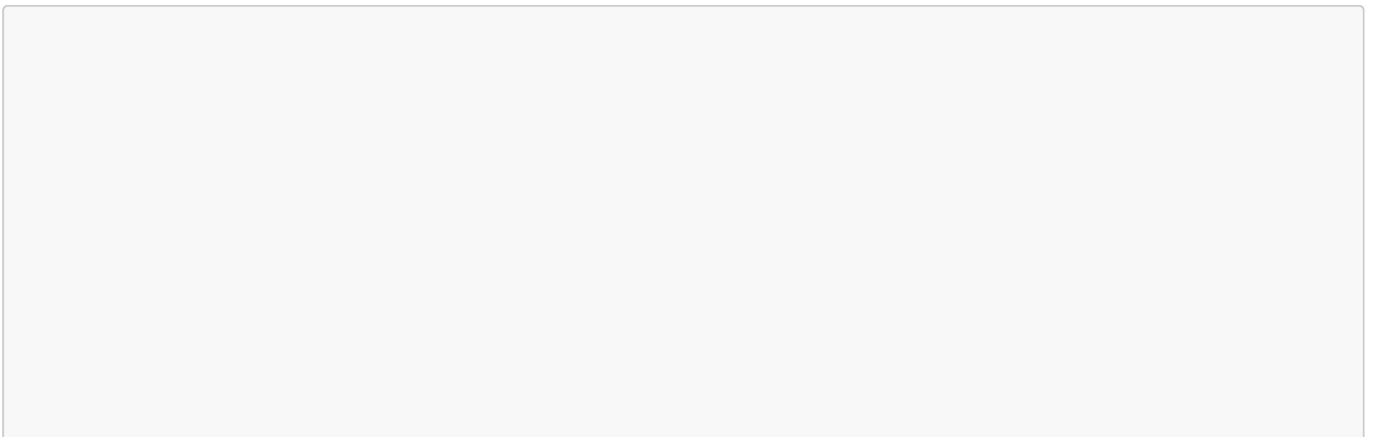
With:

- is the root directory of the service.

Define service contract



Implement service



Generate Server Stub

The tool to generate the stubs is `StubGenerator`. Use it as below:

With:

- the path of the service contract
- specify the target Stub(CLIENT/SERVER)
- Specify the language in which the stub should be generated.

Inside `src\main\resources` directory run:

It will generate the file

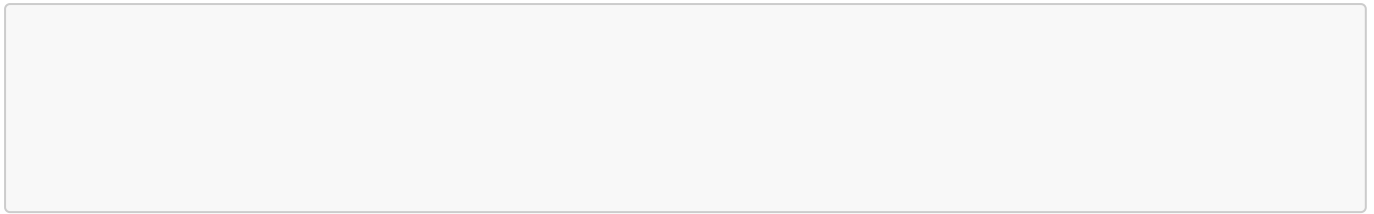
Implement the Server

With: `-Djavax.net.ssl.trustStore=src\main\resources\truststore.jks` means do not use TLS
localhost as IP address `-Dserver.ip=127.0.0.1` means the server will use the

Run the Server

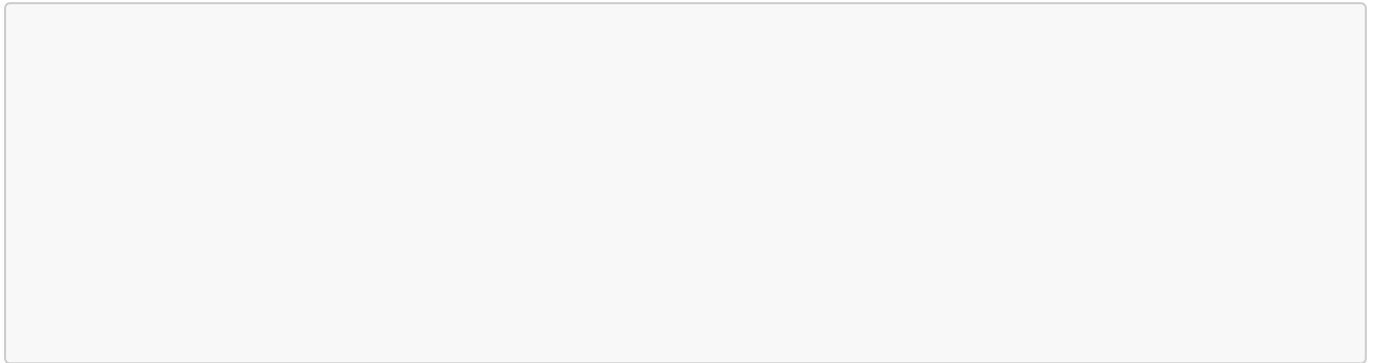
Inside `src\main\resources` directory run:

You should see something like this:



[!NOTE] By standard, SRPC uses port 5000.

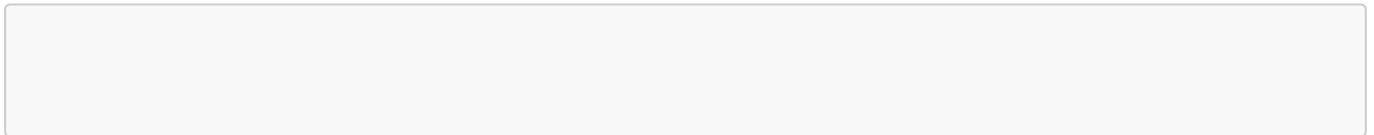
Client side directory Structure



[!NOTE] The client and server must use the same service contract. In this example, the contract is copied into both projects.

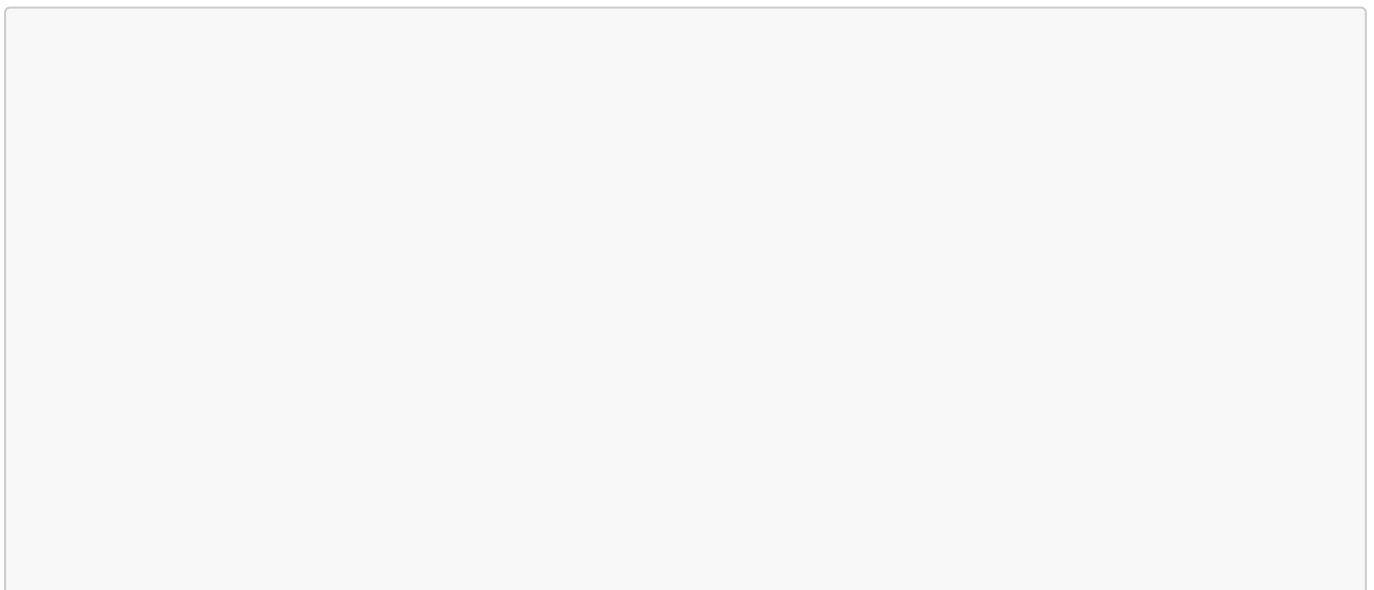
Generate Client Stub

Inside the directory run:



It will generate the file

Implement the Client



Run the Client

Component	Required pattern	Example
Service package	lowercase service name	
Service contract/interface		
Implementation script		
Interface class		
Implementation class		

Important Technical Details

- A framed application protocol with fixed-size header
- **MessagePack**-based serialization (SRPC exchanges messages as byte arrays)
- Generic error communication based on error codes
- One connection and one handler thread are used for each request.

[!NOTE] Technical documentation and release notes for version v4.x.x in progress...

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